

Amy Kuang

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Education

University of California Irvine | Master of Engineering, Mechanical & Aerospace Engineering

M.Eng. May 2027

Major Area: Autonomous and Intelligent Machines & Systems

Colorado School of Mines | Bachelor of Science, Mechanical Engineering

B.S. May 2026

Focus: Manufacturing

Technical Skills

Platforms: SolidWorks (CSWA) | LabVIEW | Arduino IDE | MATLAB | Microsoft Office | PrusaSlicer | Ideamaker | Python | CorelDraw

Technical: Lean Six-Sigma | Additive Manufacturing | Machine Shop Proficiency | Finite Element Analysis | Model-Based Systems

Personal: Team-Based Leadership | Adaptability | Attention to Detail | Cross-Functional Communication | Positive Influence | Desire to Learn | Conflict Resolution

Professional Experience

CAM Supply Inc. → Technical Product & Marketing Intern

Lake Elsinore, CA

(May 2026 – September 2026)

- Designed and 3D printed components for tattoo pens and acupuncture products using SolidWorks and IdeaMaker
- Created engineering drawings and technical documentation to support manufacturing and product use
- Designed user guides, brochures, product labels, and marketing collateral using Coreldraw
- Maintained FDA-compliant product documentation and quality records

Niagara Bottling LLC. → Manufacturing Engineer Intern

Denver, CO

(May 2025 – August 2025)

- Increased production capacity by 420,800 bottles per peak production period through pack changeover improvements
- Applied SMED principles to reduce changeover time by 12% (17 → 15 min), improving OEE and resulting in a 0.139% throughput efficiency gain
- Standardized MES fault codes and automated production tracking to support RCCA investigations
- Worked with production, maintenance, and quality teams to troubleshoot inefficiencies and develop SOPs

Field Related Projects

Embedded Electronics in Additively Manufactured Components

August 2025 – May 2026

Senior Capstone Research Project – Sponsor: Honeywell International, Inc. | Colorado School of Mines

Golden, Colorado

- Researched feasibility of embedding functional thermocouples and strain gauges within structural components using a novel multi-material Directed Energy Deposition (DED) process
- Developed a hybrid powder deposition method to safely fuse Cu and CuNi30 powders in stainless steel channels
- Translated customer requirements into ASTM-compliant test coupons, qualification plans, and final recommendations

Powder Flow Characterization for Electrodynamic DED Feeders

January 2026 – May 2026

Graduate Research Project – AMFG 501: Additive Manufacturing | Colorado School of Mines

Golden, Colorado

- Studied powder flow in DED systems using Ansys Fluent CFD and transparent-flow experiments
- Identified stagnation zones causing particle settling and feed instability
- Proposed CFD-DEM coupling after DPM particle tracking proved unstable in multi-outlet geometries

Finite Element Analysis of a Double Wishbone Suspension System

December 2024

- Redesigned and optimized lower control arm for an existing double wishbone suspension system
- Performed FEA under multiple loading conditions to validate different loading conditions and ensure reliability
- Produced a comprehensive engineering report including FEA results, cost analysis, and design iterations

PID-Controlled Easy Bake Oven

November 2024 – December 2024

- Designed and built a PID-controlled heating system using LabVIEW, thermocouple feedback, and a MOSFET control circuit
- Performed system integration, hardware-software validation, and functional testing
- Developed a custom UI to adjust PID gains and monitor system performance via live waveform charts